

FULL NAME: _____

BLOCK: _____

Logarithms + DOK-3 Spine: q15 (exp $\leftrightarrow$ log inverse via graph)

Lesson 6-3 · Quick · Teacher Edition

OBJECTIVES

Math Objective	Evaluate logarithms using the definition $\log_b x = y \Leftrightarrow b^y = x$, distinguish common (log) vs natural (ln) logs, and use the graph of $y = 3^x$ to estimate logarithmic values through the exp $\leftrightarrow$ log inverse relationship.
Language Objective	Explain the inverse relationship between exponential and logarithmic equations using “undo” language.
Essential Understanding	A logarithm answers the question: what exponent? – logs undo exponentials. If I know how to graph $y = 3^x$, I can use that same picture to estimate $\log_3 50$ because logarithms and exponentials are inverses.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Fluency evaluating logarithms; use the exp $\leftrightarrow$ log inverse relationship graphically (Topic 6 Form B prep: q15 covers Q7, Q10, Q11; q17 covers Q13).
Essential Question	If I know how to graph $y = 3^x$, how can I use that same picture to estimate $\log_3 50$ — and why does this work?
Materials	Student packet with printed $y = 3^x$ graph ($x \in [-1, 4.5]$, $y \in [0, 85]$, grid-lines). Teacher copy adds dashed $y = 50$ line and read-off annotation. [GRAPH PLACEHOLDER — needs_cleanup=YES, see Visuals Checklist].

[PIN] PRE-FLIGHT — SINGLE-PERIOD COMPRESSED LESSON + DOK-3 SPINE q15

This is a single-period quick treatment of Lesson 6-3. ONE example modeled in Launch (Ex 3: Evaluate Logarithms). DOK-3 spine q15 drives Explore.

q15: Use the graph of $y = 3^x$ to estimate $\log_3 50$. Students find $y = 50$ on the graph and read off the x-value. Since $3^3 = 27$ and $3^4 = 81$, answer is between 3 and 4, closer to 3.5. Formally $\log_3 50 \approx 3.56$. Pair-share: why does this work? (Because $\log_3 50 = x$ means $3^x = 50$ — logs and exp are inverses.)

VISUAL REQUIRED: Student packet needs printed $y = 3^x$ graph ($x \in [-1, 4.5]$, $y \in [0, 85]$, gridlines). Teacher copy: same graph + dashed $y = 50$ line + read-off annotation. See Visuals Checklist — needs_cleanup=YES.

PT#58 dropped as spine — not Form-B-aligned. q15 rehearses 3 of 8 Form B items through exp $\leftrightarrow$ log inverse relationship. The continuous-growth modeling of PT#58 will live in a future lesson if the class picks up 6-6+.

Pace: Do Now (7) $\rightarrow$ Launch (8) $\rightarrow$ Explore q15 (15 min) + q52 (4) + q53 (3) + q54 (3) $\rightarrow$ Share (5) $\rightarrow$ Exit (5). $\approx$ 48 min. Buffer in q53+q54.

45-min Wed F variant: cut q53 + q54 from Explore. NEVER cut q15.

Tag: bridge to logarithms launch

Do Now**DOK: 1–2****Minutes: 7****Teacher does**

- Hand out at door.
- Circulate 3 min.
- Cold-call 1 student on q14 error analysis.
- Pull sheets at 7 min.

Students do

- Error analysis on $e^t = 98$ (q14).
- Reason about $\log_4 x < 0$ (q16).
- Sort vocab: common vs. natural log (q3).

Questions to ask

- What is the student's error in q14? What step undoes the exponential?
- For q16: what base-4 power gives a value less than 1?
- What's the base for $\log x$ vs. $\ln x$?

Adult role

Solo teacher: circulate silently. No hints on q14 — let pairs argue.

Do Now #1 — Error Analysis (q14) (6-3-savvas-q14)

Error Analysis Describe and correct the error a student made in solving an exponential equation.

$$\begin{aligned}
 16e^t &= 98 \\
 e^t &= 6.125 \\
 6.125t &= \ln e \\
 t &= \frac{\ln e}{6.125}
 \end{aligned}$$

[CHECK] ANSWER KEY / ANTICIPATED TALK

Answer key (from source): The student did not convert to logarithmic form correctly. The solution should be $t = \ln 6.125$.

Do Now #2 — Reasoning (q16) (6-3-savvas-q16)

Generalize For what values of x is the expression $\log_4 x < 0$ true?

[CHECK] ANSWER KEY / ANTICIPATED TALK

Answer key (from source): $\{x \mid 0 < x < 1\}$

Do Now #3 — Vocabulary (q3) (6-3-savvas-q3)

Vocabulary Explain the difference between the common logarithm and the natural logarithm.

[CHECK] ANSWER KEY / ANTICIPATED TALK

(verify before class)

Tag: teacher think-aloud Example 3 (Evaluate Logarithms)

Launch

DOK: 1–2

Minutes: 8

Teacher does

- Project Example 3. Think aloud: $\log_b x = y \Leftrightarrow b^y = x$.
- Walk through $\log_5 125 = 3$ (ask: $5^? = 125$).
- Walk through $\log_{1/4} 16 = -2$ (negative exponent — flag this!).
- Walk through $\log_3 0$: Is there any power of 3 that gives 0? No – undefined.
- Walk through $\log_2 2^8 = 8$ using the identity $\log_b(b^k) = k$.
- Flag Rules card — students mark it on student packet.
- 30-sec pause: Turn to partner: rewrite $\log_4 64$ using the definition.
- Do NOT solve Practice items — release at minute 15.

Students do

- Watch Example 3 silently. Copy the $\log \leftrightarrow \exp$ switch.
- Note: $\log_b 0$ is undefined (no real answer).
- Mark the Rules card on their student packet.
- During pause: rewrite $\log_4 64$ with partner.

Questions to ask

- $\log_5 125 = ?$ — what power does 5 need?
- $\log_{1/4} 16 = ?$ — why is the answer negative?
- Why is $\log_3 0$ undefined?
- Use the identity: $\log_2 2^8 = ?$

Adult role

Project Example 3 only. Release promptly at minute 15.

[MIC] TEACHER SCRIPT

1. 'A logarithm asks: what exponent does b need to produce x?'
2. ' $\log_5 125$: what power of 5 is 125? $5^1 = 5$, $5^2 = 25$, $5^3 = 125$. Answer: 3.'
3. ' $\log_{1/4} 16$: tricky — $(1/4)^? = 16$. $(1/4)^{-2} = 4^2 = 16$. Answer: -2.'
4. ' $\log_3 0$: no power of 3 ever reaches 0 — UNDEFINED.'
5. ' $\log_2 2^8$: identity $\log_b(b^k) = k$. Answer: 8. No work needed.'
6. 'For today's DOK-3 spine: on the graph of $y = 3^x$, we can read $\log_3 50$ directly — because exp and log are inverses, the x-value IS the log.'
7. Release to Explore.

Tag: TPS + DOK-3 driver

Explore**DOK: 2–3****Minutes: 25****Teacher does**

- 4 laps across 25 min.
- Lap 1 (10 min): [STAR] q15 DOK-3 spine. Press pairs: 'Where on the graph is $y = 50$? What x-value does that correspond to?' Do NOT give the answer.
- After 10 min: pair presentations (5 min). One pair explains: 'Why does reading the x-value on the exp graph give you the log?'
- Lap 2 (4 min): q52. Press: 'Which log rule applies here?'
- Lap 3 (3 min): q53. Press conversion between log and exp forms.
- Lap 4 (3 min): q54. Press asymptote and domain reasoning.
- Then q17: press 'Is $\log_3(1/27) = -3$ correct? Use the definition to check.'
- 45-min variant: skip q53 and q54. Budget extra time on q15.

Students do

- 5 items in order: q15 $\rightarrow$ q52 $\rightarrow$ q53 $\rightarrow$ q54 $\rightarrow$ q17.
- For q15: use the printed $y = 3^x$ graph. Find $y = 50$, read the x-value.
- Pair-share for q15: explain WHY reading the x-value gives $\log_3 50$.
- q17: verify or refute $\log_3(1/27) = -3$ using the definition.
- Justify each step using sentence frame.

Questions to ask

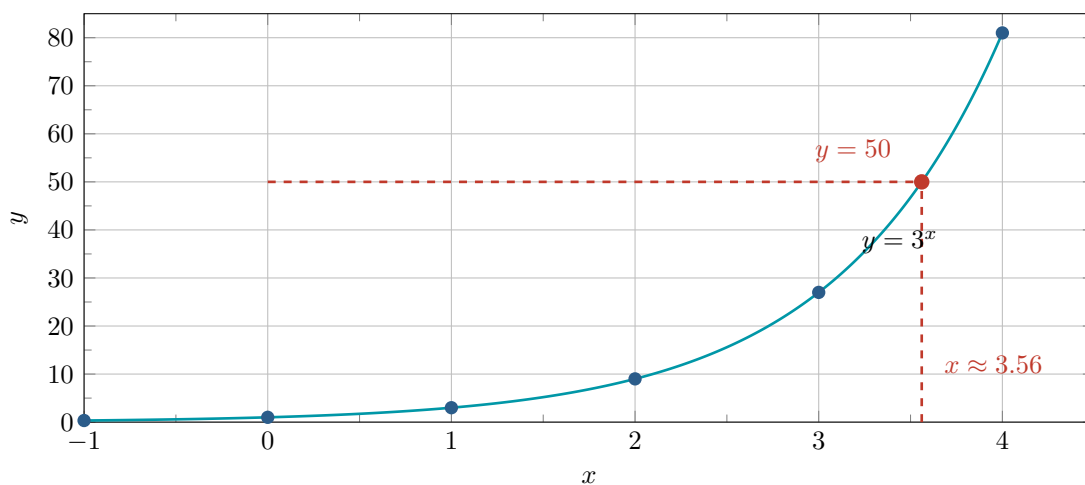
- q15: on the graph, where is $y = 50$? Between which two integer x-values?
- q15: why does the x-value where the graph hits $y = 50$ equal $\log_3 50$?
- q15: $3^3 = 27$, $3^4 = 81$ — so $\log_3 50$ is between ___ and ___?
- q17: what does $\log_3(1/27) = y$ mean in exponential form?
- q52: does the log change or the argument?
- q53: can you rewrite in exponential form first?
- q54: what is the domain of $\log_b x$? What is the asymptote?

Adult role

Solo teacher: 4 laps. On q15, press the 'why does this work?' reasoning — that's the Form B Q10/Q11 assessment-critical move.

Practice #15 [STAR] DOK-3 SPINE (6-3-savvas-q15)

Higher Order Thinking Use the graph of $y = 3^x$ to estimate the value of $\log_3 50$. Explain your reasoning.
[GRAPH / TIKZ figure]



[STAR] DOK-3 SPINE — Practice #15 (exp $\leftrightarrow$ log inverse via graph)

Prompt: Use the graph of $y = 3^x$ to estimate the value of $\log_3 50$. Explain your reasoning.

GRAPH: Student packet has $y = 3^x$ graph ($x \in [-1, 4.5]$, $y \in [0, 85]$). Teacher copy: same graph + dashed $y = 50$ line + read-off annotation. [GRAPH PLACEHOLDER — needs_cleanup=YES]

Expected reasoning: Find $y = 50$ on graph, read x-value. $3^3 = 27$, $3^4 = 81 \rightarrow$ answer between 3 and 4, closer to 3.5. Formally $\log_3 50 \approx 3.56$.

Pair-share: why does this work? Because $\log_3 50 = x$ means $3^x = 50$ — the x-value where the exp graph hits $y = 50$ IS the log.

Form B alignment: Q7 (inverse of exp), Q10 (inverse from graph), Q11 (log VA).

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): 3.5; $\log_3 50$ is an expression equivalent to $3^x = 50$, so you need to find the value of x for which $y = 50$.

Practice #52 (6-3-savvas-q52)

How long does it take for \$250 to grow to \$600 at 4% annual percentage rate compounded continuously? Round to the nearest year.

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

Practice #53 (6-3-savvas-q53)

Model with Mathematics Michael invests \$1,000 in an account that earns a 4.75% annual percentage rate compounded continuously. Peter invests \$1,200 in an account that earns a 4.25% annual percentage rate compounded continuously. Which person's account will grow to \$1,800 first?

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

Practice #54 (6-3-savvas-q54)

Reason The Richter magnitude of an earthquake is $R = 0.67 \log(0.37E) + 1.46$, where E is the energy (in kilowatt-hours) released by the earthquake.

[IMAGE: Cracked wall photo with Richter scale overlay numbered 1 through 10 and labels “Not felt” at 1 and “Extreme” near 9–10; callout text: “At a richter magnitude of 4 and above, the walls in your house may start to crack.”]

- What is the magnitude of an earthquake that releases 11,800,000,000 kilowatt-hours of energy? Round to the nearest tenth.
- How many kilowatt-hours of energy would earthquake have to release in order to be an 8.2 on the Richter scale? Round to the nearest whole number.
- What number of kilowatt-hours of energy would an earthquake have to release in order for walls to crack? Round to the nearest whole number.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. magnitude 7.9 b. 31,009,857,353 kWh c. 16,705 kWh · IMAGE PLACEHOLDER [photo]: Cracked wall photo with Richter scale overlay numbered 1 through 10 and labels “Not felt” at 1 and “Extreme” near 9–10; callout text: “At a richter magnitude of 4 and above, the walls in your house may start to crack.”

Practice #17 (promoted to core) (6-3-savvas-q17)

Use Structure A student says that $\log_3\left(\frac{1}{27}\right)$ simplifies to -3. Is the student correct? Explain.

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

[PUZZLE] FORM B ASSESSMENT ALIGNMENT

- q15 (spine) → Form B Q7 (inverse of exp), Q10 (inverse from graph), Q11 (log VA)
- q17 → Form B Q13 (condense argument reasoning)
- q52 → Form B Q9 (log↔exp definition)
- q53 → Form B Q14 (solve exp with logs, Change of Base)
- q54 → Form B Q9 + Q14 (multi-part log modeling)

Tag: academic conversation + 6-4 bridge

Share / Summary**DOK: 2****Minutes: 5****Teacher does**

- Cold-call 1 pair to present q15 reasoning using sentence frame.
- Class signals thumbs. Write on board: $\log_3 50 = x$ means $3^x = 50$.
- 6-4 bridge: ‘Next lesson: logarithmic FUNCTIONS — graphing $y = \log_b x$. Today’s log ↔ exp definition is the foundation.’

Students do

- Presenting pair explains: 'I found $y = 50$ on the graph and read $x \approx 3.56$. This works because $\log_3 50 = x$ means $3^x = 50$.'
- Rest of class signals and writes takeaway in margin.

Questions to ask

- Summarize: how does the graph of $y = 3^x$ let you find $\log_3 50$?
- Why is it that reading x gives you the log? What is the relationship?

Adult role

Cold-call a pair that got the estimation — hold them accountable to naming the $\exp \leftrightarrow \log$ inverse relationship explicitly.

Tag: summary recap (not graded)

Exit Ticket

DOK: 1–2

Minutes: 5

Teacher does

- Collect at door.
- Scan blank 3 ($e^t = 6.125 \rightarrow \ln$). Plan re-teach if many students write 'log' instead of 'ln' or leave the base blank.

Students do

- 4 sentence stems, honest.

Questions to ask

- Did students correctly state $\log_b x = y$ means $b^y = x$?
- Did students get $\log_2 32 = 5$ (since $2^5 = 32$)?
- Did students write $\ln$ (not $\log$) for base e ?

Adult role

Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

$\log_b x = y$ means ____.

$\log_2 32 = \underline{\quad\quad}$ because $2^? = 32$.

To solve $e^t = 6.125$, take the ____ of both sides giving $t = \underline{\quad\quad}$.

One biggest thing I learned today: _____.

[CLOCK] PERIOD A / PERIOD F — 50-MIN CLASS NOTES

Base lesson is 50 min. If 65 min: use extra 15 min on Explore q15 (richer pair-share on why $\exp \leftrightarrow \log$ inversion works) and q53/q54.

45-min Wed F variant: cut q53 + q54 from Explore. NEVER cut q15.

If finished early: hand out Practice #18 ($\ln 1000 \neq 3$ — fast finisher).

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Log $\leftrightarrow$ Exp Rules card as a sticker/cut-out.
- For q15, pre-label the y-axis at $y=27$, $y=50$, $y=81$ so students can locate the target values without visual confusion.
- Accept 'between 3 and 4, closer to 3.5' as full credit for the estimate.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: 'logarithm' = the exponent; 'inverse' = undoes; 'estimate' = approximate value, not exact.
- 'UNDEFINED' = no real number answer (not zero).
- Pair ELL students with English-dominant partners for TPS.